Matteo Calabretta

Catania (CT), Italy

Email • Linkedin • GitHub

Education

University of Catania – Department of Mathematics and Computer Science — Catania, Italy
M.Sc. Data Science LM-DATA GPA: 29.2/30 — Oct 2024 – present (expected 2026)

Relevant Coursework: Basics of Computing, Database & Big Data Analytics, Statistical Learning, Optimization, Data Protection Law, Industrial Data Science.

University of Catania – Department of Economics — Catania, Italy
B.Sc. Economics L-33 Final grade: 90/110 — Sep 2019 – Dec 2023

Relevant Coursework: Statistics, Mathematical Finance, International Relations, Macroeconomic Policy, Commercial Law.

Charles University – Faculty of Mathematics and Physics — Prague, Czech Republic
Erasmus+ Exchange Student — M.Sc. Computer Science programme — Feb 2026 – Jun 2026

Relevant Coursework: Data Engineering, Statistical Methods

Academic Projects

Heatwave & Mortality Analysis

- Built an end-to-end ETL pipeline (Eurostat, Open-Meteo) analysing temperature effects on weekly mortality across 59 EU NUTS2 regions.
- Designed a star-schema data model with atomic parquet writes, config-driven extraction, and coverage-aware missing-data handling.
- Applied epidemiological methods (age standardization, relative risk, confounding) to quantify heat-mortality associations.
- Delivered an interactive Streamlit dashboard with maps, region comparison, and lag/season controls over shared analysis logic.

Predictive Modelling Project – Statistical Learning Course

- Applied supervised learning algorithms (regression, decision trees, ensemble methods) to a real-world tabular dataset; conducted cross-validation and hyperparameter tuning.
- Produced a written report comparing model performance metrics (RMSE, AUC, F1) and communicated findings with interpretability tools (feature importance, SHAP values).

Smart-Factory Case Study – Data Science in the Factory of the Future

- Analysed industrial sensor data streams from a simulated smart-factory environment; developed anomaly-detection and predictive-maintenance models.
- Integrated data-acquisition logic with visualisation dashboards, delivering actionable KPI summaries for a mock management audience.
- Developed from start to end a machine learning pipeline on Azure.

Skills & Interests

Technical: Python (pandas, scikit-learn, NumPy, Matplotlib, PyTorch), SQL, R, Git/GitHub, LaTeX, Docker, Apache Airflow, Azure ML.
Languages: Italian (native), English (Cambridge certificate B2 level)
Areas of Interest: Machine Learning, Data Engineering, Data-Driven approaches.
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[1] I authorize the processing of my personal data contained in my CV, in accordance with EU Regulation 2016/679 (GDPR).